

Business Insight Analysis Report

Customer Segmentation & Sales Performance (RFM Analysis)

Dataset	Online Retail Transactions (E-commerce, UK-based retailer)
Period Covered	December 2010 – December 2011
Tools Used	Python — pandas, numpy, matplotlib
Analysis Type	RFM (Recency, Frequency, Monetary) Customer Segmentation
Prepared For	Business Stakeholders / Marketing & Sales Team

1. Executive Summary

This report analyzes transaction-level sales data from an online retail business to understand revenue trends and customer purchasing behavior. Using an RFM (Recency, Frequency, Monetary) framework, the customer base was segmented into five actionable groups. The analysis identifies which customers drive the majority of revenue, which are at risk of churning, and where marketing and retention efforts should be focused.

Total Revenue	\$7.86M
Unique Customers	2,997
Total Transactions (lines)	97,826
Analysis Period	13 months
Top Segment by Revenue	Champions (Best Customers) – 61.3% of sales

Key Takeaways

- A small group of highly engaged customers (Champions) generates the large majority of revenue — classic Pareto (80/20) behavior in retail.
- A meaningful share of the customer base has gone quiet (Lost / Hibernating) and represents a revenue-recovery opportunity through win-back campaigns.
- Revenue is seasonal, with a strong pre-holiday spike, meaning inventory and staffing should be planned around that peak.
- Revenue is heavily concentrated in one country, indicating both a stronghold to defend and an opportunity to diversify internationally.

2. Methodology

The analysis was performed in Python using **pandas** for data loading, cleaning and aggregation, **numpy** for numerical scoring calculations, and **matplotlib** for visualization. A full step-by-step technical walkthrough is provided in the companion *Technical Documentation* report.

- **Data cleaning:** removed rows with missing Customer ID, and removed cancelled / negative-quantity orders and zero-price line items.
- **Feature engineering:** calculated $TotalAmount = Quantity \times UnitPrice$ for every line item.
- **RFM calculation:** for every customer, computed Recency (days since last purchase), Frequency (number of distinct invoices), and Monetary (total amount spent).
- **Scoring:** each of Recency, Frequency and Monetary was split into 5 quantile-based bands (1–5) using `pandas.qcut`, then averaged into a single RFM score.
- **Segmentation:** customers were grouped into five business-friendly segments (Champions, Loyal Customers, At Risk / Can't Lose, Recent New Customers, Lost / Hibernating) based on their Recency and Frequency scores.

3. Monthly Revenue Trend

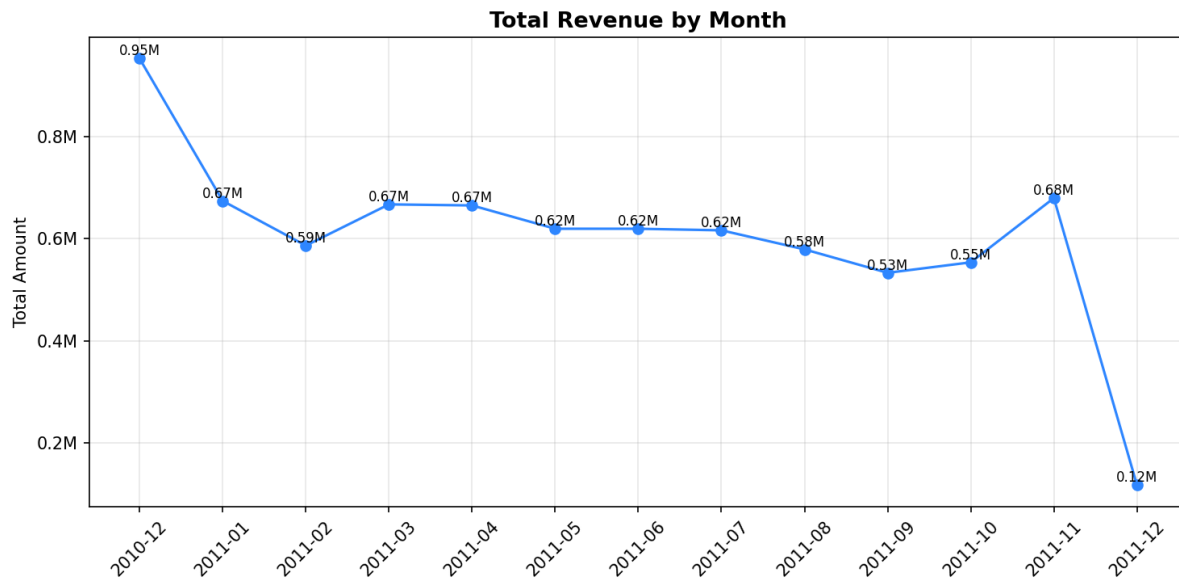


Figure 1: Total revenue by month, December 2010 – December 2011.

Monthly revenue fluctuates between roughly \$0.5M and \$0.7M for most of the year, with a clear seasonal uplift heading into November as customers begin holiday shopping. December 2011 appears artificially low because the dataset's transaction history ends on December 8, 2011, so that month is only partially captured. **Business implication:** inventory, staffing and marketing spend should be scaled up ahead of the November peak to avoid stockouts and to capture the seasonal demand surge.

4. Revenue Share by Customer Segment

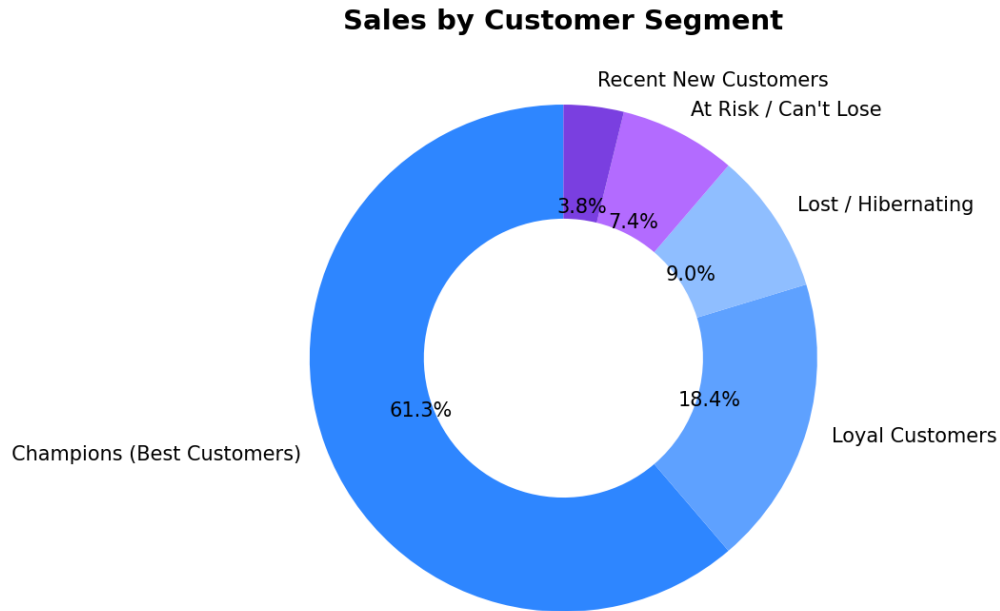


Figure 2: Share of total revenue contributed by each RFM customer segment.

Champions — the most recent and frequent buyers — contribute 61.3% of total revenue despite being a minority of the customer base. Loyal Customers add a further 18.4%. Together, these two segments account for nearly 80% of revenue, reinforcing the classic 80/20 pattern seen in most retail businesses.

Business implication: retention programs (loyalty perks, early access, personalized offers) aimed at Champions and Loyal Customers protect the majority of the revenue base and typically offer the best return on marketing spend.

5. Customer Engagement by Segment (Avg RFM Score)

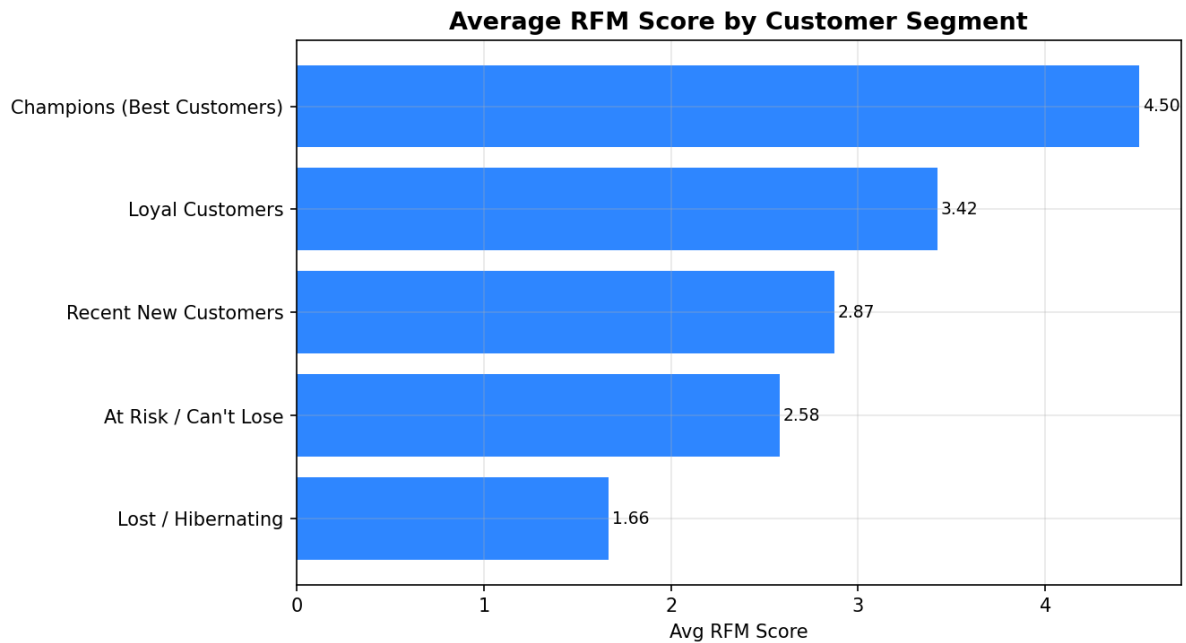


Figure 3: Average RFM score for each customer segment (higher = more engaged).

The average RFM score confirms a clear engagement hierarchy: Champions (4.50) score highest, followed by Loyal Customers (3.42), Recent New Customers (2.87), At Risk / Can't Lose (2.58), and Lost / Hibernating (1.66) score lowest. **Business implication:** the At Risk and Lost segments together represent customers who used to buy (or bought once) but have since gone quiet — a natural target list for re-engagement emails, discount codes, or 'we miss you' campaigns before they are lost permanently.

6. Total Revenue by Segment

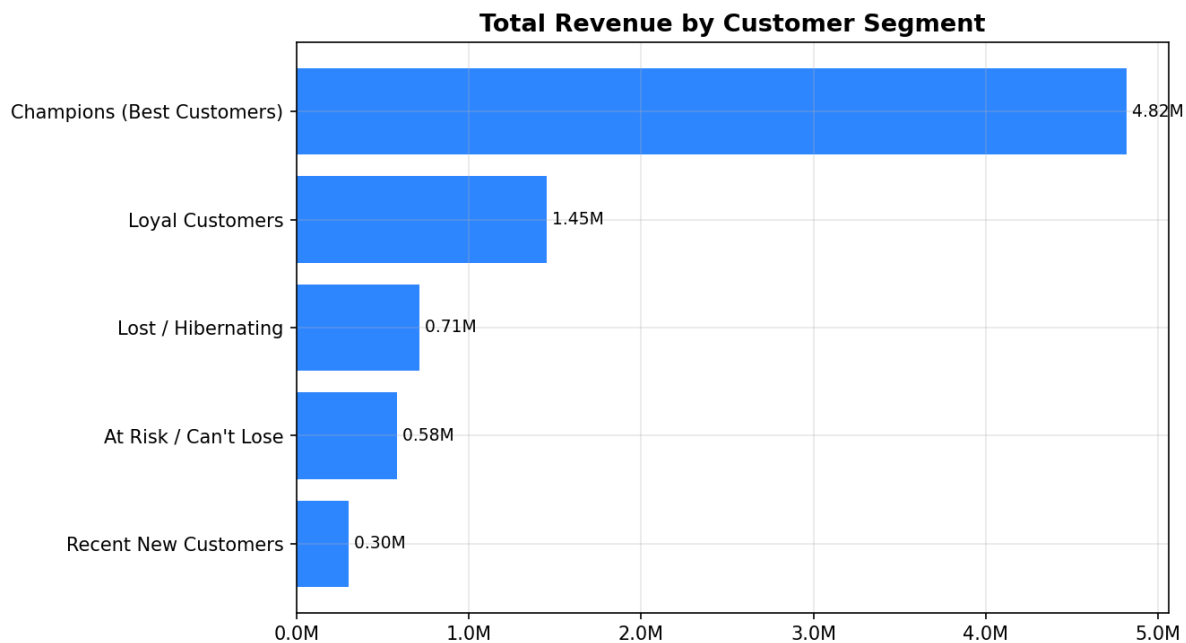


Figure 4: Total revenue generated by each customer segment.

Segment	Customers	Total Sales	Avg RFM Score
Champions (Best Customers)	807	\$4,818,210.45	4.50
Loyal Customers	602	\$1,449,093.35	3.42
Lost / Hibernating	948	\$710,854.87	1.66
At Risk / Can't Lose	389	\$584,903.49	2.58
Recent New Customers	251	\$301,146.19	2.87

Interestingly, Champions are fewer in number than Lost / Hibernating customers (807 vs. 948) but generate almost 7x more revenue. This highlights that customer *value* is far more concentrated than customer *count*. **Business implication:** revenue-protection strategy should prioritize the health of the Champions segment above all else — losing even a handful of these customers has an outsized impact on total revenue.

7. Geographic & Product Insights

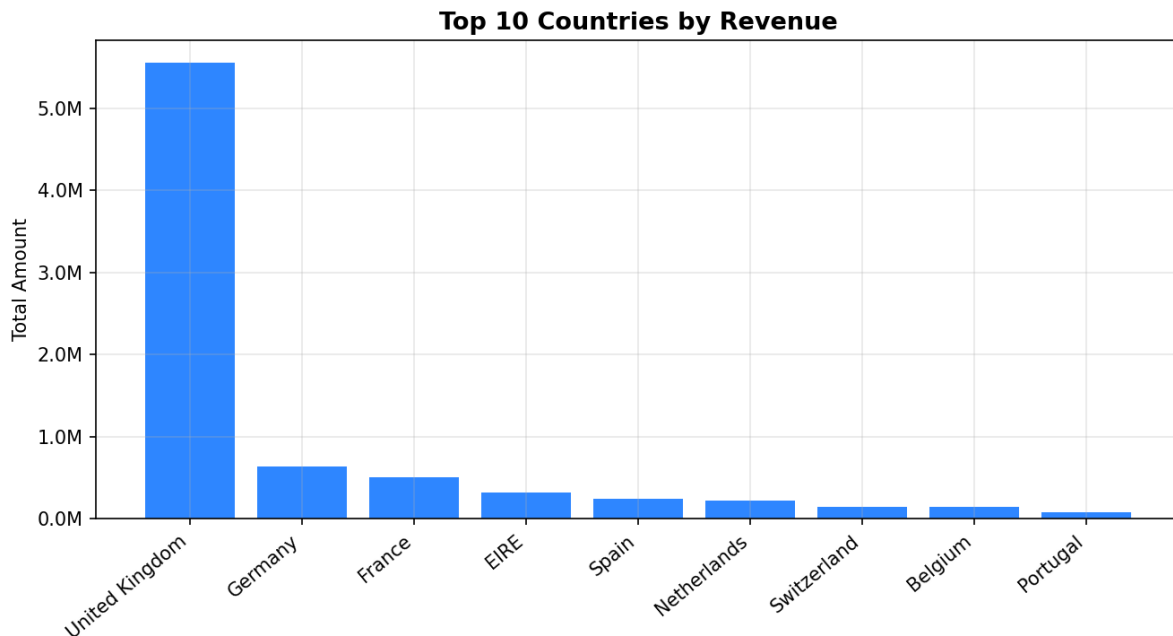


Figure 5: Top 10 countries by total revenue.

The United Kingdom is overwhelmingly the largest market, generating roughly 70% of total revenue, followed at a distance by Germany and France. **Business implication:** the business is heavily reliant on a single domestic market; targeted expansion into the next-largest markets (Germany, France, EIRE) could diversify revenue and reduce geographic concentration risk.

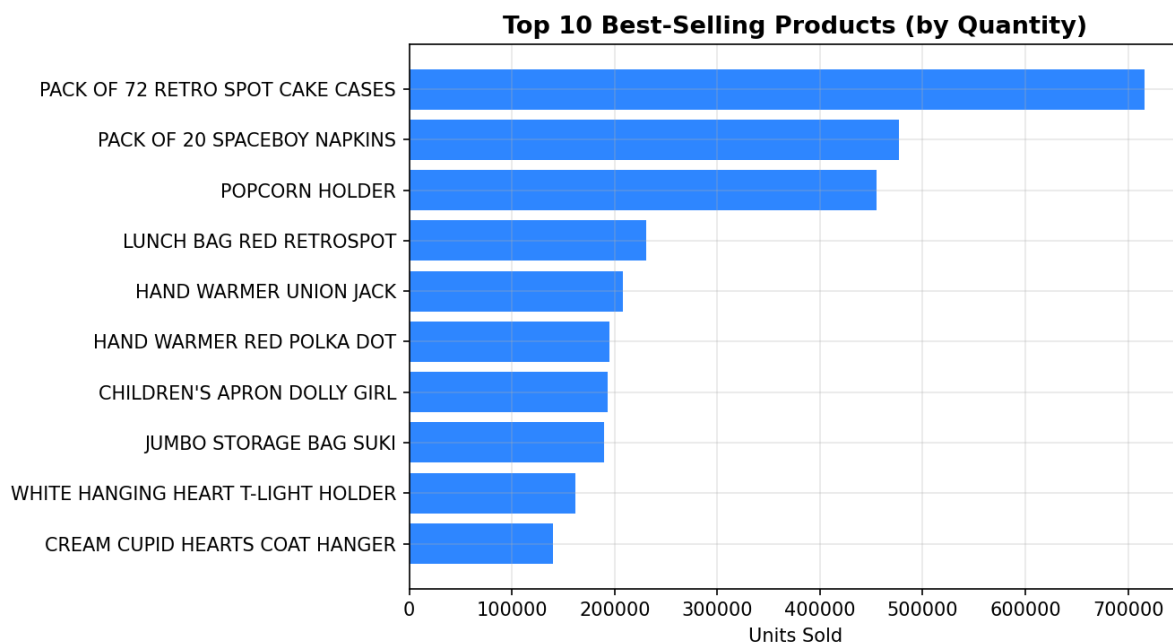


Figure 6: Top 10 best-selling products by units sold.

Low-cost, high-volume novelty and gift items (cake cases, napkins, popcorn holders, lunch bags) dominate unit sales. These act as frequent, low-friction purchases that keep customers coming back. **Business implication:** these products are strong candidates for cross-sell and bundle promotions, and should be prioritized for stock availability given their high turnover.

8. Recommendations

- **Protect Champions:** introduce a VIP/loyalty tier with early access, free shipping, or exclusive bundles to reduce churn risk among the top 27% of customers who drive 61% of revenue.
- **Win back Lost / Hibernating customers:** launch a re-engagement email campaign with a time-limited discount; even a modest reactivation rate would be meaningful given this is the largest segment by customer count.
- **Nurture Recent New Customers:** set up an onboarding sequence (welcome offer, product recommendations) in the first 30–60 days to convert them into Loyal Customers.
- **Plan for seasonality:** increase inventory and marketing budget ahead of the November peak.
- **Diversify geographically:** test targeted campaigns in Germany and France, the next-largest markets after the UK.

Note: This report is based on transaction data processed for illustrative business-analysis purposes, following the same analytical workflow (data cleaning → RFM scoring → segmentation → visualization) demonstrated in the source Jupyter notebook, "Customer Segmentation Analysis."